FinTrust Data Quality Check

Data Quality Check	Customer Dataset	Transaction Dataset	Observations / Business Impact
Number of Records	1,500	12,000	The datasets provide sufficient records for customer and transaction analysis. The Transaction dataset is substantially larger and provides detailed transaction-level information.
Number of Columns	12	11	Both datasets have a manageable number of fields and are within the expected structural scope. No structural concerns were identified.
Missing Values	0	192	The Customer dataset has no missing values. The Transaction dataset contains 192 missing values: 96 in **Device_Type** and 96 in **Location**. These gaps may affect device- and location-based segmentation and analysis.
Duplicate Records	0	0	No duplicate records were identified. This supports the reliability of record-level analysis and reduces the risk of double-counting.
Incorrect / Inconsistent Values	0	0	No obvious inconsistent formatting, invalid categorical entries, or incorrect values were identified during the initial check.
Data Type Issues	None identified	None identified	Numeric fields and date/time fields were verified to be stored in appropriate formats. No major data-type conversion was required.
Potential Outliers	1	1,474	One potential outlier was identified in **Digital_Engagement_Score**. The Transaction dataset contains 1,474 potential outliers in **Amount_NGN**. These may reflect the right-skewed nature of transaction values rather than data-entry errors and should therefore be investigated before removal.
Relationship Between Datasets	—	—	All Transaction records reference a valid **Customer_ID** in the Customer dataset. This confirms referential integrity and indicates that there are no orphaned transaction records.

Overall Data Quality Assessment

The initial data quality assessment indicates that the FinTrust datasets are **generally suitable for analysis**. There are no duplicate records, no confirmed incorrect or inconsistent values, and no major data-type issues.

The main data-quality considerations are the **192 missing values in the Transaction dataset** and the **potential outliers in transaction amounts**. The missing Device_Type and Location values should be handled carefully before conducting detailed channel, device or location analysis. The transaction amount outliers should not automatically be removed because large transactions may represent genuine customer activity.

The successful relationship between the two datasets also allows customer-level information, such as **Customer_Segment, Account_Type and Digital_Engagement_Score**, to be combined with transaction-level information for deeper analysis.

Recommended Actions

1. Replace missing Device_Type and Location values with **"Unknown"** where appropriate, rather than deleting the affected transactions.
2. Investigate the 1,474 potential Amount_NGN outliers to determine whether they represent genuine high-value transactions.
3. Investigate the one potential Digital_Engagement_Score outlier.
4. Retain valid high-value transactions unless there is evidence that they are data-entry errors.
5. Use the validated Customer_ID relationship to perform customer-segment and transaction-behaviour analysis.